



Stock Tracker

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User Stories

- For adults who want a paper-trading platform where they can buy, sell, and keep track of trade, stock, and personal balances.
“I want to get into stock trading”



Features / Requirements Document

Core Features -

User login/authentication

Add/Remove a trade(Buy and Sell)

View portfolio summary

View portfolio positions and their values

Cash balance updates

Deposit and Withdraw money

Error handling

Functional Requirements:

Fetch a stock's price using the Massive API

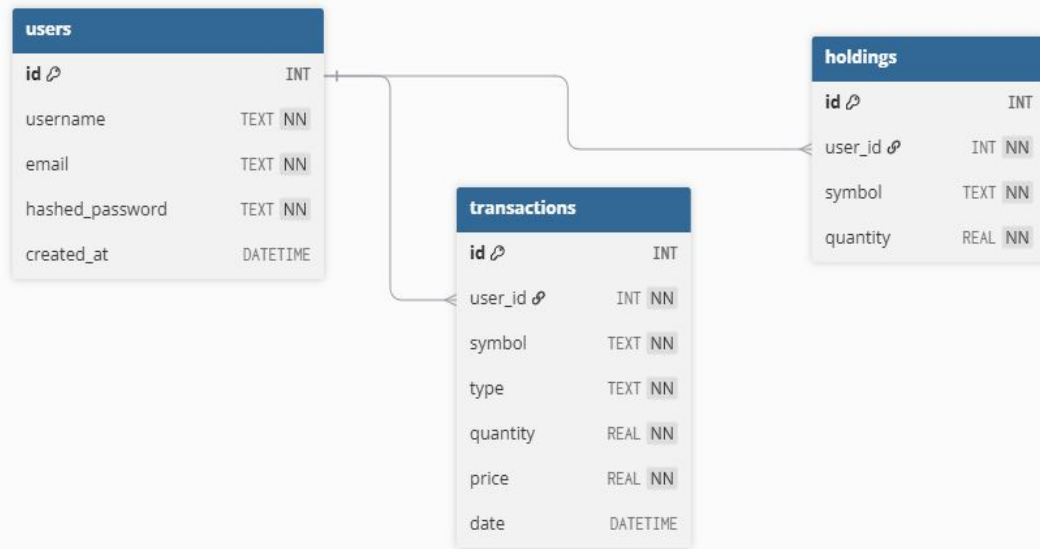
Subtract cash on BUY and add cash on SELL

Update share quantity and position value

Each BUY/SELL must update the database

Database Diagram

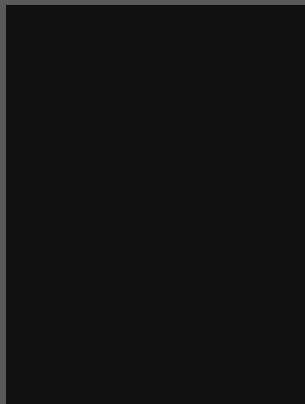
```
1 Table users {
2   id INT [pk, increment]
3   username TEXT [unique, not null]
4   email TEXT [unique, not null]
5   hashed_password TEXT [not null]
6   created_at DATETIME
7 }
8
9 Table transactions {
10  id INT [pk, increment]
11  user_id INT [not null, ref: > users.id]
12  symbol TEXT [not null]
13  type TEXT [not null]
14  quantity REAL [not null]
15  price REAL [not null]
16  date DATETIME
17 }
18
19 Table holdings {
20  id INT [pk, increment]
21  user_id INT [not null, ref: > users.id]
22  symbol TEXT [not null]
23  quantity REAL [not null]
24 }
25
```



Style Guide

Primary

#111111



Accent

#22c55e



Secondary

#2d2d2d



Text Box Bg

#111827



Box Border

#374151



Environment Setup

Front End:

Jinja2 templates with HTMX dynamic updates

ChartJS and Tailwind for styling

Database:

SQLite3 with SQLAlchemy ORM

Back End:

Python

FastAPI web framework

Uvicorn for running server and handling HTTP requests

Tools:

Git/Github for version control

VS Code as text editor

Bruno for API testing

Discord for communication

Docs



[Project 3 Research Log.xlsx](#)